DAA EXPERIMENT 9

Index of First Occurrence in a String

Aim

To find the index of the first occurrence of a given character in a string.

Algorithm

1. Read a string and the character to search for.
2. Initialize the index to -1.
3. Traverse the string from left to right.
4. If the current character matches the search character, store its index and stop searching.
5. If no match is found, the index remains -1.
6. Display the index.
7. Stop.

Program

```
# Index of First Occurrence in a String

s = input("Enter a string: ")
ch = input("Enter character to search: ")

index = -1

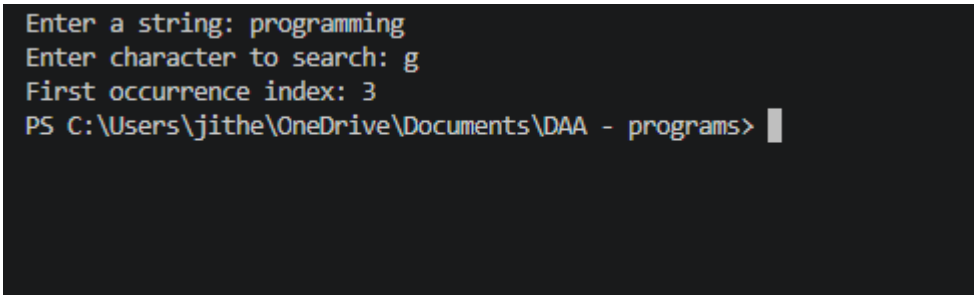
for i in range(len(s)):
    if s[i] == ch:
        index = i
        break

print("First occurrence index:", index)
```

Complexity Analysis

- Best Case: O(1)
- Average Case: O(n)
- Worst Case: O(n)
- Space Complexity: O(1)

Output



Result

Thus, the index of the first occurrence of the specified character was successfully found.